Diagram bank: what varies between process families

Use any of these visuals in the final pitch to make the 15-family OOD story concrete.

same vocabulary

different valid routes

held-out families

Litho levels

How many mask / photoresist patterning cycles the route needs: 4, 5, or 6.

Extra via block

Some families repeat field oxide, via etch, via fill, and CMP.

Spacer rate

How often spacer dielectric and anisotropic spacer etch appear.

Implant route

Valid route recombinations, e.g. source/drain followed by N-buffer.

Pitch thesis

A family is not only a label. It is a different valid manufacturing route over the same process-step vocabulary. Holding out families therefore tests route logic instead of recipe memorization.

Family variation matrix

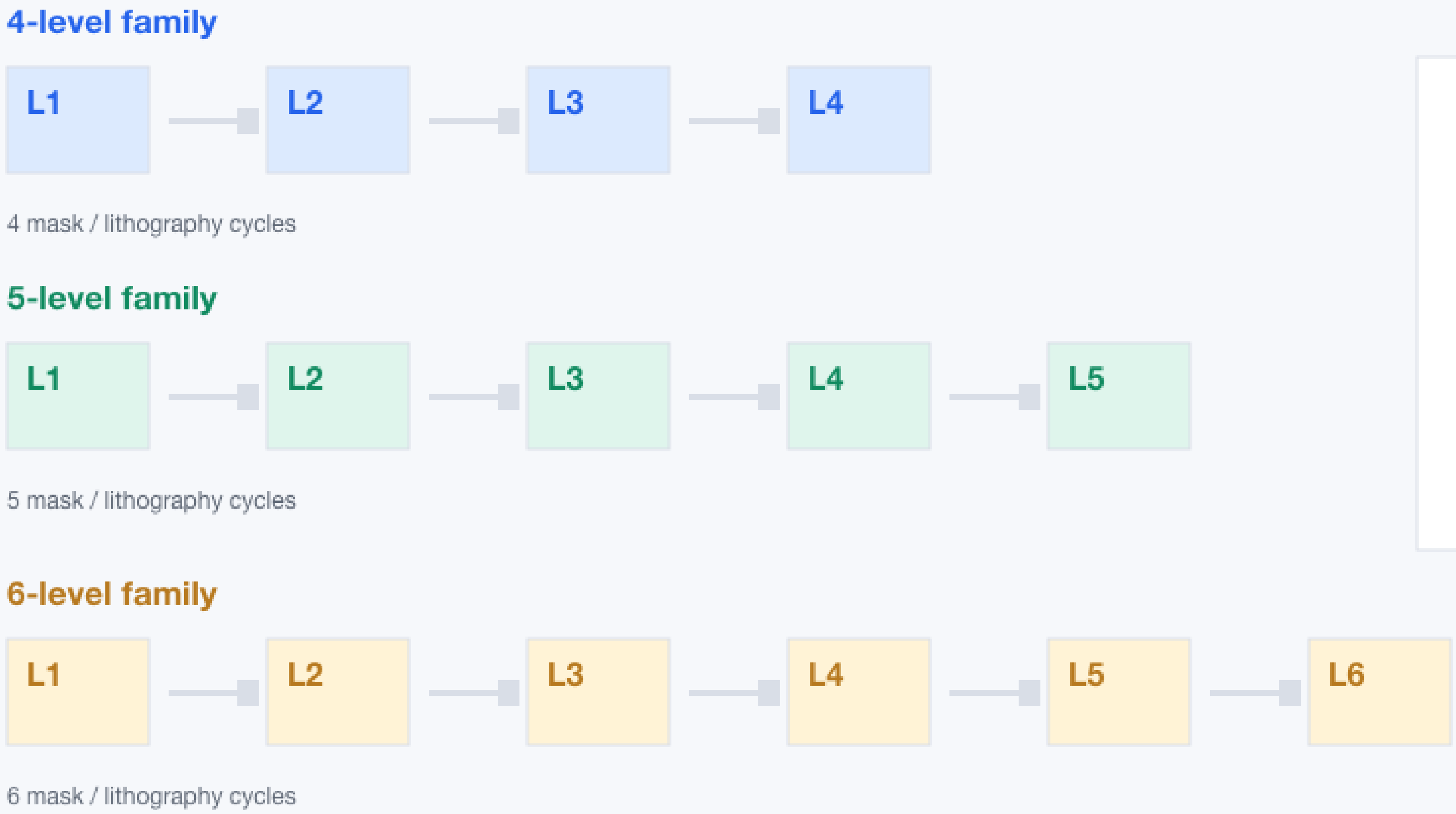
A compact table to explain the experimental design in one slide.

Group	Litho levels	Extra via block	Spacer rate	Role	Interpretation
MOSFET	4	native	native	train	Known organizer route
IGBT	6	native	native	train	Longer known route
IC	4	native	native	train	Different known route
SCFAM01-09	4-6	mixed	0.25-0.80	train	Teaches route diversity
SCFAM10-12	4-6	mixed	0.20-0.90	unseen test	Hidden-family proxy

The important control: no new step tokens. The model sees familiar process words in unfamiliar valid structures.

Litho levels = repeated mask patterning cycles

More litho levels mean deeper process routes and more opportunities for valid optional branches.

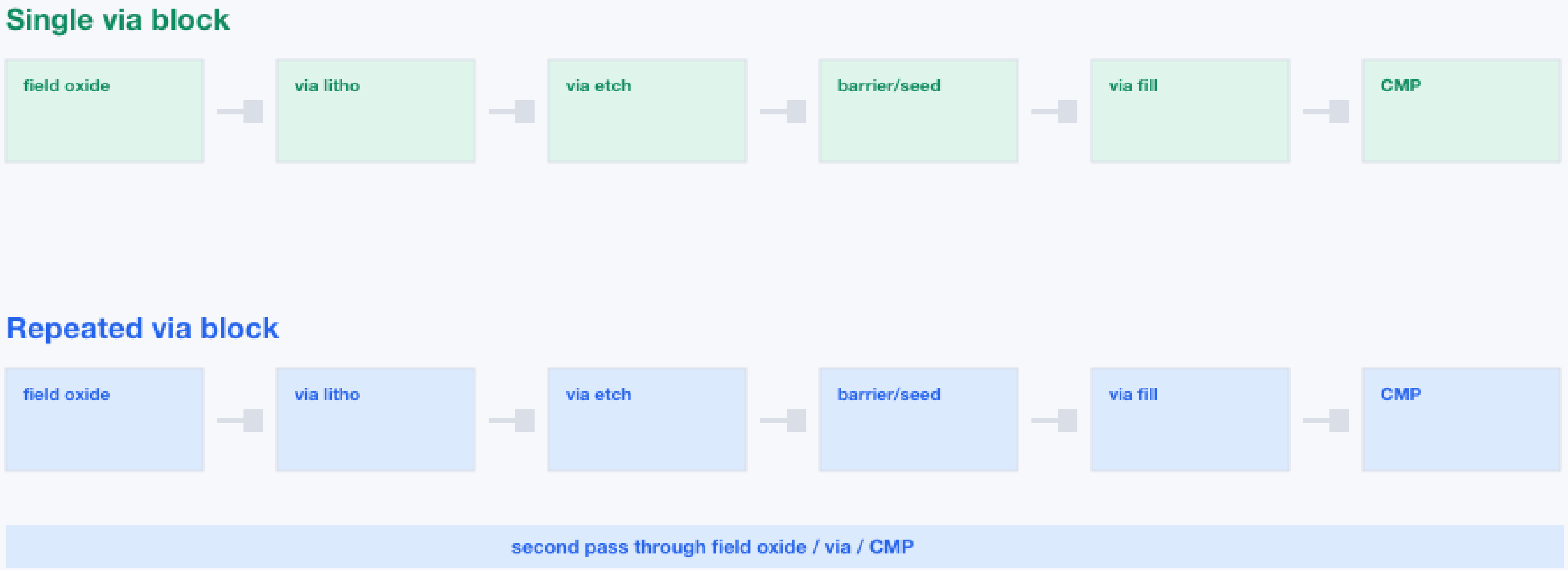


One litho cycle

1. spin resist
2. align mask
3. expose
4. develop
5. inspect
6. etch / implant
7. strip + clean

Extra via block = a repeated interconnect sub-route

Some families include one via stack; others repeat it, changing valid order and sequence length.

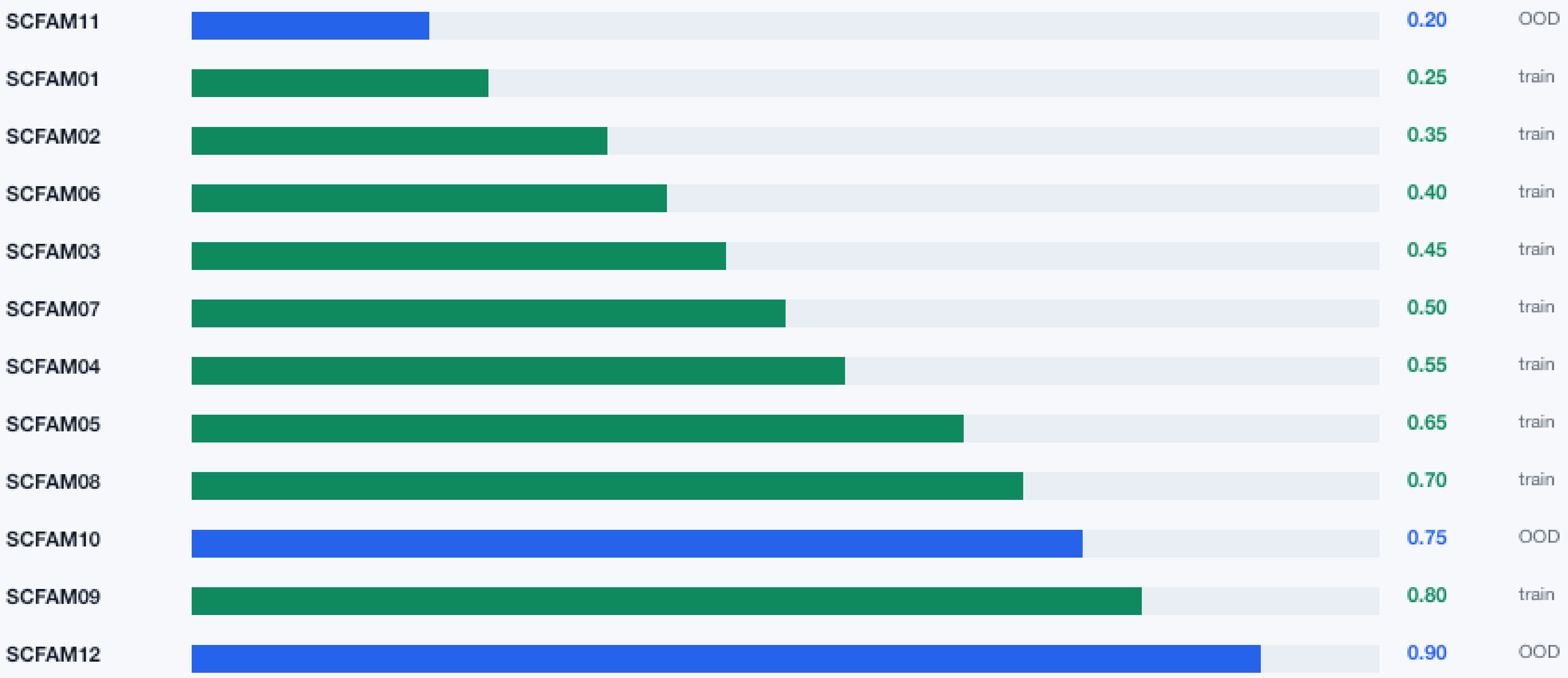


Why it matters: the next correct step depends on whether the family is entering, repeating, or leaving this sub-route.

Spacer rate = optional block frequency

The spacer block is valid but optional. Families differ by how often it appears.

Probability that spacer dielectric + anisotropic spacer etch are inserted



Why this is OOD

Held-out families sit at low, medium-high, and high spacer rates. The model must handle optional blocks without seeing these exact family profiles.

Implant route = valid recombination of process logic

Synthetic families combine known implant logic in new valid routes.

MOSFET-like: poly etch → source / drain → anneal → spacer

IGBT-like: poly etch → channel stop → N-buffer later → anneal

Synthetic: poly etch → source / drain → N-buffer same cycle → anneal

The vocabulary is known, but the transition SOURCE/DRAIN -> N-BUFFER in the same cycle is a new route pattern.

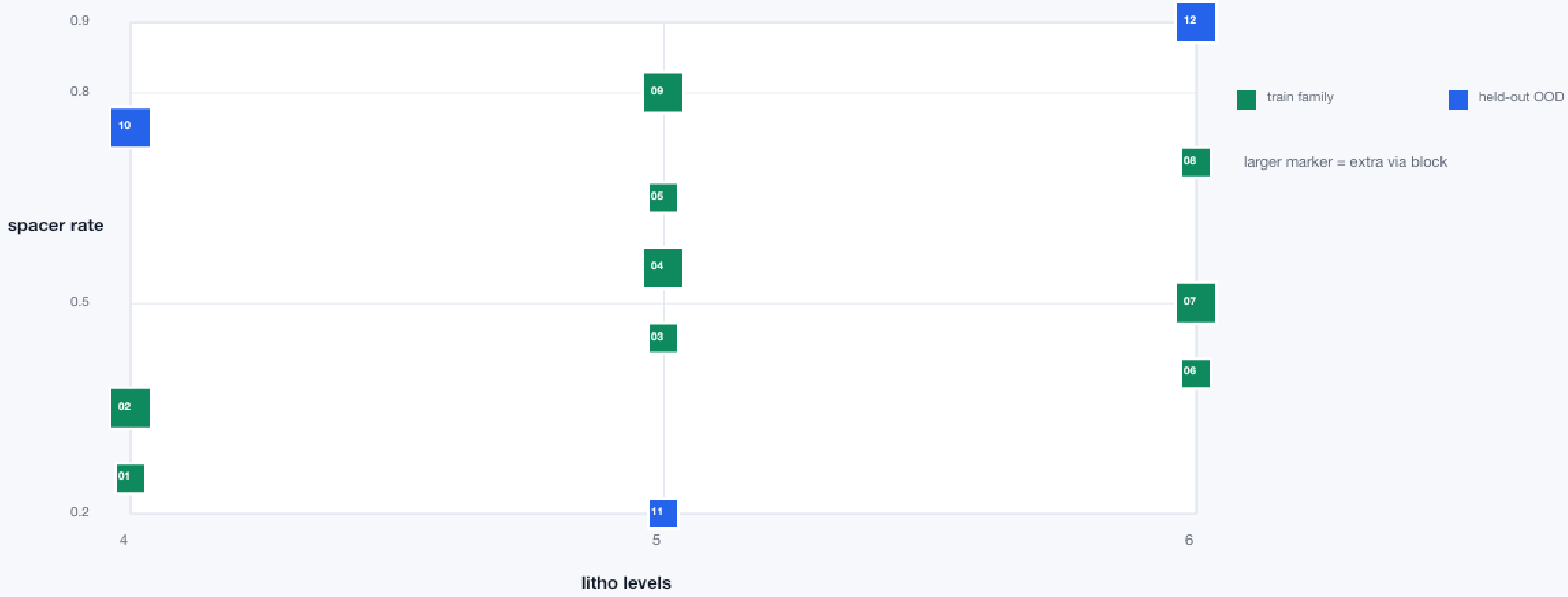
Feature heatmap across synthetic families

This is the concrete family profile used by the 12-train / 3-held-out protocol.

Family	Split	Litho	2nd via	Spacer
SCFAM01	train	4	no	0.25
SCFAM02	train	4	yes	0.35
SCFAM03	train	5	no	0.45
SCFAM04	train	5	yes	0.55
SCFAM05	train	5	no	0.65
SCFAM06	train	6	no	0.40
SCFAM07	train	6	yes	0.50
SCFAM08	train	6	no	0.70
SCFAM09	train	5	yes	0.80
SCFAM10	OOD	4	yes	0.75
SCFAM11	OOD	5	no	0.20
SCFAM12	OOD	6	yes	0.90

Feature-space view of held-out families

The OOD families are withheld profiles, not random rows from the same profile.



Same tokens, different valid order

The model is not asked to invent new process words. It must learn route order.

Shared process vocabulary

clean | oxidize | litho | etch | implant | anneal | metal | test

Known route

clean → oxidize → litho → implant → anneal → metal

OOD route

clean → oxidize → litho → implant → implant → via repeat

This is why fixed-vocabulary OOD is fair: failures reflect process-order generalization, not unknown-token handling.

Copyable pitch explanation

A short verbal bridge for whichever diagram you choose.

One-sentence version

A process family is a route profile: it changes lithography depth, repeated via blocks, spacer frequency, and implant order while keeping the same process vocabulary.

20-second version

We did not just add more rows. We generated family profiles that vary real process structure: 4/5/6 lithography levels, optional second via stacks, different spacer probabilities, and implant-route recombinations. Then we trained on 12 families and tested on 3 unseen family profiles. That makes the benchmark closer to the hidden-family question.

Defensive wording

The synthetic families reuse the existing vocabulary and are validated by the organizer rule checker, so the OOD test measures route logic rather than new-token handling.